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ELECTRIC MOTOR COOLING SYSTEM

Abstract

Systems and methods for a sealing assembly for a motor cooling system are herein described. In one example, the sealing assembly includes an elastomeric seal interposed radially between a lateral surface of a stator of an electric motor and a sealing ring. The elastomeric seal includes a plurality of openings configured to receive a plurality of end windings of the stator and a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate a coolant for the plurality of end windings from a rotor.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an electric motor with a cooling system that includes a sealed cavity which is formed around a stator end winding.

BACKGROUND AND SUMMARY

[0002] In electric vehicle (EV) motor applications, stator winding cooling has been used in an attempt to achieve greater motor efficiency. For instance, efficiency targets may demand the flow of oil across the stator, in certain applications. However, in certain prior cooling system designs, oil is directed through interior cavities from which oil can leak into an air gap between the rotor and the stator. Oil in the air gap results in drag losses which results in a significant decrease in motor efficiency. The inventors have recognized that sealing the stator cooling arrangement off from other regions of the motor, such as the rotor cavity, may avoid an undesirable drop in motor efficiency.

[0003] One approach for a motor cooling system includes a sealing ring and a sealing sleeve. In the motor cooling system, a cavity is formed between the sealing ring and the sealing sleeve in which a stator end winding is at least partially immersed in a coolant (e.g., oil). Interfaces between the sealing ring, sealing sleeve, stator, and the rotor cavity may be sealed to reduce flow of coolant from the coolant channels to the rotor cavity. For example, some interfaces may be sealed by one or more of a gasket, an O-ring, or adhesive sealant. However, sealing a potential leak path to the air gap formed at an interface between an axial wall of the rotor cavity and the interface between the sealing ring and the stator is particularly challenging due to minimal surface area available for positioning a seal. Seal protrusion at positions past the interface and into the rotor cavity may degrade the seal, increasing risk of oil leakage from the coolant channels into the rotor cavity. Adhesive sealants may not be desirable due to chemical incompatibility with the coolant.

[0004] To overcome at least some of the abovementioned issues the inventors developed a sealing assembly for an electric motor cooling system. In one example, the sealing assembly comprises an elastomeric seal interposed radially between a lateral surface of a stator of an electric motor and a sealing ring, where the elastomeric seal comprises a plurality of openings configured to receive a plurality of end windings of the stator and a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate a coolant for the plurality of end windings from a rotor. In this way, a sealed cavity is provided that prevents fluid flow into the rotor cavity, and specifically to the motor's air gap. Consequently, the motor may achieve a target efficiency, if so desired.

[0005] In another example, the molded annular bead may include in an uncompressed state a sealing surface having a concave profile in cross section. The molded annular bead may include a first protrusion arranged on a first face of the elastomeric seal, a second protrusion arranged on a second face of the elastomeric seal, and a concave groove arranged axially between the first protrusion and the second protrusion. The first protrusion may ramp up and inward toward the inner diameter of the elastomeric seal, and the second protrusion may ramp down and inward toward the inner diameter of the elastomeric seal. The concave groove may include a first edge, a second edge, and groove surface extending axially therebetween. The concave groove may include the first edge positioned approximately parallel with a first planar surface of the elastomeric seal and the second edge positioned approximately parallel with a second planar surface of the elastomeric seal. In another example, upon final compression in an electric motor assembly, the molded annular bead may extend up to, and no further than, an axial wall of the sealing ring in a radially inward direction. In this way, the geometry of the molded annular bead maintains the elastomeric seal on the minimal surface area, fluidly sealing the rotor cavity from coolant without the elastomeric seal protruding into the rotor cavity.

[0006] It should be understood that the summary above is provided to introduce in simplified form a selection of concepts that are further described in the detailed description. It is not meant to identify key or essential features of the claimed subject matter, the scope of which is defined

uniquely by the claims that follow the detailed description. Furthermore, the claimed subject matter is not limited to implementations that solve any disadvantages noted above or in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0007] FIG. 1 shows an electric motor with a cooling system.

[0008] FIG. 2 shows a cross-sectional view of the electric motor and cooling system depicted in FIG. 1.

[0009] FIG. 3 shows an example electric motor with a sealing ring, sealing sleeve, and example of an elastomeric seal.

[0010] FIG. 4 shows an underside view of the elastomeric seal and sealing ring depicted in FIG. 3.

[0011] FIG. 5 shows an enlarged view of the underside view of the elastomeric seal and sealing ring depicted in FIG. 4.

[0012] FIGS. 6A and 6B show different views of the elastomeric seal depicted in FIG. 3.

[0013] FIG. 7 shows an enlarged view of the example electric motor depicted in FIG. 3.

DETAILED DESCRIPTION

[0014] Systems and methods for a sealing assembly for a motor cooling system are described herein. The sealing assembly includes an elastomeric seal, a sealing ring and a sealing sleeve that are compressively retained in a motor housing to form a sealed cavity around stator end windings. The sealed cavities enable coolant (e.g., oil) to be directed through the stator and a plurality of end windings of the stator while preventing the coolant from entering a rotor cavity. In particular, the elastomeric seal includes a molded annular bead that is compressed between a sealing land of the stator and the sealing ring to create a seal. The seal created via the molded annular bead may at least fluidly seal the rotor cavity at the sealing land and the sealing ring. The geometry of the molded annular bead profile compresses in a manner that does not protrude into a rotor cavity of the electric motor and allows for constant sealing pressure and contact width without over-stressing the elastomeric material. The elastomeric seal may have high compliancy, sealing the rotor cavity across a range of temperature and pressure conditions. Consequently, the sealing ring and sleeve may be securely retained within the motor with the elastomeric seal, thereby reducing the likelihood of coolant leakage from the sealed cavity. As a result, drag losses caused by coolant in the air gap can be avoided.

[0015] FIG. 1 shows an illustration of an electric motor **100**. The electric motor **100** may be designed as an electric motor-generator and may be included in a system **102** which may take a variety forms. For instance, the electric motor **100** may be incorporated into an electric drive system of an electric vehicle (EV), in one example. As such, the electric motor is a traction motor and the electric drive may further include a transmission (e.g., gearbox), for instance. In the EV example, the EV may be an all-electric vehicle (e.g., a battery electric vehicle (BEV)), in one example, or a hybrid electric vehicle (HEV) with an internal combustion engine, in another example. However, the motor may be used in other suitable systems (e.g., stationary systems), in other examples, such as in industrial machines, agricultural systems, mining systems, and the like.

[0016] The electric motor **100** includes a rotor **104** that electromagnetically interacts with a stator **106** to drive rotation of a rotor shaft **108** that is included in the rotor. The electric motor **100** in the illustrated example includes a housing **110** with an electrical interface **112** for the stator **106**. The electrical interface **112** may be a multi-phase electrical interface with multiple electrical connectors **114**. The electrical interface **112** is a three-phase interface, in the illustrated example. However, it will be understood that the electrical interface may be a six phase interface or a nine phase interface, in other examples. More generally, the electric motor **100** may be a multi-phase

alternating current (AC) machine. However, in other examples, the electric motor **100** may be a direct current (DC) machine.

[0017] As illustrated in FIG. **1**, the electric motor **100** may be electrically coupled to an inverter **116**. The inverter **116** is designed to convert direct current (DC) power to alternating current (AC) power and vice versa. As such, the electric motor **100** may be an AC electric motor, as indicated above. However, in other examples, the electric motor **100** may be a DC electric motor (as previously indicated) and the inverter **116** may therefore be omitted from the system **102**. The inverter **116** may receive electric energy from one or more energy storage device(s) **118** (e.g., traction batteries, capacitors, combinations thereof, and the like). Arrows **120** signify the electric energy transfer between the electric motor **100**, the inverter **116**, and the energy storage device(s) **118** that may occur during different modes of system operation.

[0018] The system **102** may additionally include a control sub-system **180** with a controller **182**. The controller **182** includes a processor **184** and memory **186**. The memory **186** may hold instructions stored therein that when executed by the processor **184** cause the controller **182** to perform the various methods, control techniques, and the like, described herein. The processor **184** may include a microprocessor unit and/or other types of circuits. The memory **186** may include known data storage mediums such as random access memory, read-only memory, keep alive memory, combinations thereof, and the like.

[0019] The controller **182** may receive various signals from sensors **188** positioned in different locations in the system **102**. The sensors **188** may include an electric machine speed sensor, energy storage device temperature sensor(s), an energy storage device state of charge sensor(s), an inverter power sensor, and the like. The controller **182** may also send control signals to various actuators **190** coupled at different locations in the system **102**. For instance, the controller may send signals to the inverter **116** to adjust the rotational speed of the electric motor **100**. In another example, the controller **182** may send a command signal to the electric motor **100** and/or the inverter **116** and in response motor speed may be adjusted. The other controllable components in the system **102** may function in a similar manner with regard to command signals and actuator adjustment.

[0020] The system **102** may also include one or more input device(s) **192** (e.g., an accelerator pedal, a brake pedal, a console instrument panel, a touch interface, a touch panel, a keyboard, combinations thereof, and the like). The input device(s) **192**, responsive to user input, may generate a motor speed adjustment request.

[0021] An axis system is provided in FIG. **1**, as well as FIGS. **2-7**, for reference. The z-axis may be a vertical axis (e.g., parallel to a gravitational axis), the x-axis may be a lateral axis (e.g., horizontal axis), and/or the y-axis may be a longitudinal axis, in one example. However, the axes may have other orientations, in other examples. Rotational axis **199** of the electric motor **100** is further provided for reference in FIG. **1** as well as FIGS. **2-7**. A cutting plane **2-2** for the cross-sectional view depicted in FIG. **2** is provided in FIG. **1**. The cutting plane **2-2** extends through the rotational axis of the electric motor **100**.

[0022] Features described as axial may be approximately parallel with an axis referenced unless otherwise specified. Features described as counter-axial may be approximately perpendicular to the axis referenced unless otherwise specified. Features described as radial may circumferentially surround or extend outward from an axis, such as the axis referenced, or a component or feature described prior as being radial to a referenced axis, unless otherwise specified. Unless otherwise specified, the axis referenced may be the rotational axis **199**.

[0023] FIG. **2** shows a cross-sectional view of the electric motor **100** and the cooling system **200** for the motor. The rotor **104** and the stator **106** of the electric motor **100** are again depicted along with the housing **110** that at least partially encloses the rotor and the stator. The rotor shaft **108** is further illustrated in FIG. **2**.

[0024] The cooling system **200** may include a pump **202** and a filter **204** may uptake and deliver a coolant (e.g., oil such as a natural and/or synthetic oil) into the stator **106**. The pump **202** may

create a pressure differential by which the pump **202** may uptake and pressurize coolant. The filter **204** may remove particulates and other impurities from the coolant. Arrows **205** depict the flow of coolant between the pump **202**, the filter **204**, and the cooling system **200**. To elaborate, the pump **202** may deliver coolant to coolant passages **206** that axially traverse a stator core **208** of the stator **106**, in one example. However, other coolant flow patterns may be used in other examples. The pump **202** and the filter **204** are schematically depicted. However, it will be understood that they may have greater complexity, in practice. Further, the pump and the filter are shown spaced away from the electric motor **100**.

[0025] An air gap **210** is formed between a rotor core **212** of the rotor **104** and the stator core **208**. Due to the sealing of the coolant in the cooling system **200** expanded upon herein, the likelihood of coolant entering the air gap **210** is substantially reduced (e.g., avoided).

[0026] The cooling system **200** further includes a first sealing ring **214** and a first sealing sleeve **216** positioned on a first axial side **218** of the stator **106** (e.g., the stator core **208**). The first axial side **218** may specifically be a weld side, in one example. The first sealing ring **214** may be coupled to the stator **106** (e.g., the stator core). However, in other examples, the first sealing ring may be machined or otherwise integrally formed in the stator **106**.

[0027] The first sealing sleeve **216** and the first sealing ring **214** are coupled via a first sealing interface **222** and a second sealing interface **224**. The first sealing sleeve **216** and the stator **106** meet at a first lateral sealing interface **284**. The first sealing interface **222**, the second sealing interface **224**, and the first lateral sealing interface **284** form a sealed cavity **226** in which the stator end winding **228** are positioned. The sealed cavity **226** may receive and/or deliver coolant to/from the coolant passages **206** in the stator core **208**. For example, the sealed cavity **226** may at least partially immerse the stator end windings at in coolant. The sealed cavity **226** is fluidly separated from a rotor cavity **227**. In this way, coolant entering the air gap may be avoided, thereby increasing motor efficiency.

[0028] The first sealing interface **222** may be formed between the housing **110** and an extension **230** of the first sealing sleeve **216** that is positioned radially inward from the end windings **228**. To elaborate, the first sealing interface **222** may include a recess **231** profiled to receive a seal **233** such as an O-ring, a gasket, a diamond seal, and/or another type of liquid seal. The extension **230** of the first sealing sleeve **216** may taper in an axially inward direction (to increase the strength of the sleeve when compared to a shoulder formed as a thinner wall

[0029] The first sealing ring **214** includes a flange **232** that axially extends outward from the first axial side **218** of the stator **106**. The flange **232** seals against an interior surface **234** of the first sealing sleeve **216** to form the second sealing interface **224**. The second sealing interface **224** may specifically be a radial sealing interface. Therefore, at the second sealing interface **224**, the flange **232** and the interior surface **234** of the first sealing sleeve **216** enable a strong seal to be achieved. However, it is to be appreciated that other types of sealing interfaces may be used in other examples. The interior surface **234** of the first sealing sleeve **216** may include recesses sized to receive seals **236** such as O-rings, grommets, or a liquid seal.

[0030] The second sealing interface **224** may be positioned radially inward from the stator end winding **228** but radially outward from the first sealing interface **222**, in one example. In this way, the sealed cavity **226** may be securely sealed to provide a coolant enclosure for the end windings **228**. Coolant **239** circulates around the cavity and specifically through the end windings **228**, in this way increasing stator cooling when compared to systems that solely direct coolant around the end windings **228**.

[0031] The first lateral sealing interface **284** may be a radial seal formed between a lateral surface **282** of the stator **106** and the flange **232** of the first sealing ring **214**. For example, the first lateral sealing interface **284** may be positioned axially inward from the first axial side **218** of the stator **106**. In other words, the first lateral sealing interface **284** may be positioned to the right of the second sealing interface **224** along the y-axis. The first lateral sealing interface **284** may include a

gasket, a diamond seal, a liquid seal, and/or an elastomeric seal **220** of the present disclosure. The elastomeric seal **220** is represented schematically.

[0032] In one example, the elastomeric seal **220** may be interposed radially, with respect to the rotational axis **199**, between the first sealing ring **214** and the lateral surface **282** of the stator **106**, positioned adjacent to the air gap **210**. The elastomeric seal **220** may include openings configured to receive the end windings **228**. The elastomeric seal **220** may extend radially from an inner circumferential section **286** of the first sealing ring **214** to an outer circumferential section **280** of the first sealing ring **214**, the inner circumferential section **286** facing the rotor core **212** and the outer circumferential section **280** facing the housing **110**. The first lateral sealing interface **284** including the elastomeric seal **220** further separates the coolant **239** from the air gap **210**. In this way, the sealed cavity **226** is fluidly isolated from the rotor cavity **227**. Examples of the elastomeric seal **220** are described in more detail with reference to FIGS. 3-7.

[0033] The housing **110** may be formed in different sections that are coupled to one another, in one example. For instance, a crown side section **240** and a weld side section **242** may be coupled to a housing body **244**. The housing body **244** may circumferentially enclose the stator **106** and sealing sleeves and rings in the cooling system **200**. Fasteners **246** and/or other suitable attachment devices may be used to attach the crown side section **240** and/or the weld side section **242** to the housing body **244**.

[0034] The first sealing sleeve **216** may include an inner radial wall **248**, an outer radial wall **250**, and an axial wall **252** that allow for the sealed cavity **226** to enclose the end windings **228**. However, other contours of the first sealing sleeve **216** may be used in other examples. For instance, the first sealing sleeve may include curved sections that enclose the end windings.

[0035] An end **251** of the outer radial wall **250** may be adjacent to or in face sharing contact with a surface **253** of an outer circumferential section **280**. When the first sealing ring **214** is attached to the first sealing sleeve **216**, the outer radial wall **250** of the first sealing sleeve **216** may be adjacent to or in contact with the outer circumferential section **280** and may overlap an outer flange (not shown). In this way, the upper section of the sealed cavity **226** may be created.

[0036] The first sealing sleeve **216** in the second sealing interface **224** includes a shoulder **254** that allows for the axial compression of the sealing assembly to be tuned. To elaborate, the shoulder **254** hard mounts to the first sealing ring **214**, when the sealing assembly is installed in the housing **110** under compression. Arrows **256** indicate the compressive force exerted on the assembly formed between the first sealing ring **214** and the first sealing sleeve **216**. In this way, the first sealing ring and the sleeve may be effectively maintained in a desired position. Due to this compressive attachment of the sealing ring and sleeve in the housing, the use of attachment devices for securing the ring and/or sleeve to the housing may be avoided, if desired. Consequently, manufacturing efficiency may be increased in relation to systems which utilize fasteners. A second sealing ring **260** and a second sealing sleeve **262**, described in greater detail herein, may be compressed in a similar manner.

[0037] The coolant passages **206** may extend through the stator core **208** from the first axial side **218** to a second axial side **258**. In this way, coolant axially traverses the stator to enable heat to be effectively removed therefrom. The motor can consequently achieve greater efficiency, if so desired, when compared to prior motor cooling systems.

[0038] A second sealing ring **260** and a second sealing sleeve **262** are positioned on the second axial side **258** (e.g., a crown side) of the stator **106**, in the illustrated example. However, in other examples, the second sealing ring and sleeve may be omitted from the motor cooling system.

[0039] Similar to the first sealing ring **214** and the first sealing sleeve **216**, the second sealing ring **260** and the second sealing sleeve **262** may form a third sealing interface **264** and a fourth sealing interface **266**. The second sealing ring **260** and the stator **106** may form a second lateral sealing interface **288**. Again, the third sealing interface **264** may be a face seal formed between the housing **110** and an extension **267**. The fourth sealing interface **266** may be a radial seal formed between a

flange **269** of the second sealing ring **260**. The second lateral sealing interface **288** may be a radial seal formed between a lateral surface **290** of the stator **106** and the flange **269** of the second sealing ring **260**. The second sealing ring **260** and the second sealing sleeve **262** may be differently sized from the first sealing ring **214** and the first sealing sleeve **216** to accommodate for the varied sizes of the weld and crown side end windings. However, the second sealing ring **260** and the second sealing sleeve **262** may include similar structural features to those included in the first sealing ring **214** and the first sealing sleeve **216**. However, as expanded upon herein the second sealing sleeve may be contoured to accommodate for attachment between the electrical interface **112** and the stator end winding **228**.

[0040] An elastomeric seal **271** may be interposed radially between the lateral surface **290** of the stator core **208** and the second sealing ring **260**, in one example. Similar to the elastomeric seal **220**, the elastomeric seal **271** may further separate coolant for the end windings from the rotor core **212**. However, in another example, the second sealing ring may be adhesively attached to the stator via an adhesive. Seals **277** may be included in the fourth sealing interface **266** and a seal **279** may be included in the third sealing interface **264**. The seal **279** may include one or more of an O-ring, a gasket, a diamond seal, and a liquid seal. Further, the seals **277** may include one or more O-rings, grommets, and/or liquid seals.

[0041] The second sealing sleeve **262** may include an inner radial wall **268**, an outer radial wall **270**, and an axial wall **272** that allow for a sealed cavity **273** to enclose the end windings **228**. However, other contours of the second sealing sleeve **262** may be used in other examples.

[0042] The second sealing sleeve **262** again includes a shoulder **274** that has axially compressive force transmitted therethrough to enable the axial compression of the sealing assembly formed between the second sealing sleeve **262** and the second sealing ring **260** to be tuned as desired. This axial compression is represented via arrows **275**.

[0043] Turning to FIG. 3, a sealing assembly **300** of a cooling system of an electric motor is depicted, which may be embodiments of the cooling system **200** and electric motor **100** of FIGS. 1 and 2, respectively. The sealing assembly **300** of the cooling system may include an elastomeric seal **306** interposed radially between a lateral surface **302a** of a stator **302** and a sealing ring **304**. The lateral surface **302a** may be adjacent to or in contact with the sealing ring **304** and the elastomeric seal **306**, the elastomeric seal **306** being compressed axially against both of the lateral surface **302a** and the sealing ring **304**. A sealing sleeve **308** may circumferentially surround the sealing ring **304**, forming a sealed cavity **310** therebetween. The elastomeric seal **306** is in fluid communication with the sealed cavity **310**. The elastomeric seal **306** may be an embodiment of the elastomeric seals **220**, **271** described above with respect to FIG. 2. A box 7-7 for the detail view depicted in FIG. 7 is provided in FIG. 2. The box 7-7 depicts a sealing interface formed between the sealing ring **304**, the lateral surface **302a**, and the elastomeric seal **306**.

[0044] The sealing ring **304** may include an inner flange **312** and outer flange **314**. The sealing sleeve **308** may include an inner radial wall **316** and an outer radial wall **318**. The outer radial wall **318** of the sealing sleeve **308** may be adjacent to or in contact with the outer flange **314** of the sealing ring **304**. The inner radial wall **316** may be adjacent to or in contact with the inner flange **312**. As described above with reference to FIG. 2, there may be one or more of an O-ring, a gasket, a diamond seal, and a liquid seal interposed between the sealing ring **304** and the sealing sleeve **308** at one or more sealing interfaces formed there between, such as sealing interface **320**. In this way, the sealing ring **304** and the sealing sleeve **308** create the sealed cavity **310**.

[0045] The inner flange **312** may include a first axial wall **322** and the stator **302** may include a second axial wall **324**, the first axial wall **322** and the second axial wall **324** both extending in parallel axially with respect to the rotational axis **199**. A sealing interface **326** is formed between the first axial wall **322** and the second axial wall **324**. In one example, the sealing interface **326** comprises a sealing land having minimal available area for contact and sealing with a gasket. In one example, the sealing land may be approximately 2 mm. The elastomeric seal **306** is configured

to seal the sealing interface **326**. In particular, the elastomeric seal **306** includes a molded annular bead **328** arranged to define an inner diameter of the elastomeric seal **306**. The molded annular bead **328** is configured to be compressed axially, thereby sealing the sealing interface **326**. In this way, the elastomeric seal **306** further separates a coolant enclosed in the sealed cavity **310** and circulating through the cooling system from a rotor (e.g., rotor core **212** in FIG. 2).

[0046] FIG. 4 shows an underside view **400** of the sealing ring **304** and the elastomeric seal **306** of the sealing assembly **300** of a cooling system of an electric motor. The elastomeric seal **306** is shown in an uncompressed state. Elements introduced with reference to FIG. 3 are numbered similarly and not reintroduced. A box 5-5 for the enlarged view depicted in FIG. 5 is provided in FIG. 4.

[0047] In one example, the elastomeric seal **306** may include an inner circumference **412**, an outer circumference **414**, an outer diameter **416**, and an inner diameter **418**. The elastomeric seal **306** may be substantially annular (e.g., ring-shaped) including a central opening **420**. The elastomeric seal **306** may be radially symmetric relative a central axis that is coaxial with the rotational axis **199**. The elastomeric seal **306** may be substantially planar (e.g., flat), including a planar portion **410** extending radially from the molded annular bead **328** to the outer diameter **416**. The elastomeric seal **306** includes a plurality of openings including a plurality of square openings **404** arranged radially outward from a plurality of rectangular openings **402**. The plurality of openings may be arranged in the planar portion **410** and surrounding the central opening **420**. In one example, each square opening of the plurality of square openings **404** may be radially aligned with a rectangular opening of the plurality of rectangular openings **402** with respect to the rotational axis **199**. To elaborate, each of square opening of the plurality of square openings **404** and each of rectangular opening of the plurality of rectangular openings **402** may be sectioned symmetrically by a common axis or line that intersects the rotational axis **199**.

[0048] The sealing ring **304** may include an inner circumference **422** and an outer circumference **424** that respectively align with the inner circumference **412** and the outer circumference **414** of the elastomeric seal **306**. To elaborate, the inner circumference **422** and the inner circumference **412** super impose and the outer circumference **424** and the outer circumference super impose **414** when the sealing ring **304** contacts the elastomeric seal **306**. The sealing ring **304** may include a plurality of channels, including a plurality of first channels **406** arranged radially outward from a plurality of second channels **408**. The plurality of first channels **406** and the plurality of second channels **408** may respectively align with the plurality of rectangular openings **402** and the plurality of square openings **404**. To elaborate, the plurality of first channels **406** align directly over the corresponding rectangular openings **402**. Similarly, the plurality of second channels **408** superimpose with the square openings **404**. The first channels **406** may be formed between baffles **426** that extend in radial directions and are aligned with a stator (e.g., stator **106** in FIGS. 1-2) such that coolant passes through the first channels **406** and is directed towards the end windings. The first channels **406** may have a rectangular cross-sectional contour, in one example. However, other channel profiles have been contemplated.

[0049] The plurality of rectangular openings **402** are shaped to receive the plurality of end windings of a stator (e.g., end windings **228** of stator **106** in FIG. 2). The plurality of square openings **404** are configured to be in fluid communication with the plurality of rectangular openings **402** and the cooling system (e.g., cooling system **200** in FIG. 2). For example, the elastomeric seal **306** may be arranged over the end windings against the lateral surface **302a** of the stator **302** (e.g., see FIG. 3). The end windings may pass through the plurality of rectangular openings **402**, through the plurality of first channels **406**, into the sealed cavity (e.g., sealed cavity **310** in FIG. 3). Coolant may enter through the plurality of rectangular openings **402** into the sealed cavity via the first channels **406** where the coolant may exchange heat with the stator windings, and exit the sealed cavity through the plurality of second channels **408** aligned with the square openings **404**.

[0050] In one example, the elastomeric seal **306** may be relatively more flexible, more compressible, and more compliant than the sealing ring **304**, the sealing sleeve **308**, or the stator **302** (see FIG. 3). In one example, the elastomeric seal **306** may be formed from molded rubber and the plurality of rectangular openings **402** and the plurality of square openings **404** may be die cut. Additionally, or alternatively, the elastomeric seal **306** may be formed from acrylic, sponge rubber, ceramic paper, or polyurethane. The sealing ring **304** and the sealing sleeve **308** may be formed from relatively more rigid materials, such as metal, ceramic, or composites of metal or ceramic. In this way, at least some portions of the geometry of the elastomeric seal **306** may change when compressed axially between the sealing ring **304**, the stator **302** and the sealing sleeve **308**. The geometry of the elastomeric seal, and how at least some of the geometry may change under various pressure and heat conditions is described in more detail below.

[0051] FIG. 5 shows an enlarged view **500** of the underside view **400** of the sealing ring **304** and the elastomeric seal **306** depicted in FIG. 4. Elements introduced with reference to FIGS. 3-4 are numbered similarly and not reintroduced.

[0052] The enlarged view **500** shows the plurality of rectangular openings **402** and the plurality of square openings **404** are radially arranged in the planar portion **410** of the elastomeric seal **306**. The plurality of rectangular openings **402** and the plurality of square openings **404** may be radially arranged with respect to the rotational axis **199**. In one example, each rectangular opening of the plurality of rectangular openings **402** may be defined by an opening on a first face **510** of the elastomeric seal **306**, an opening on an opposing, second face **610** (e.g., see FIG. 6A-6B), and a rectangular surface **508**. Each rectangular opening of the plurality of rectangular openings **402** may include pair of long sides **506** and a pair of short sides **504** arranged perpendicular to the pair of long sides **506**. The plurality of rectangular openings **402** may include a first short side **504a** and a second short side **504b**. The first short side **504a** of each of the plurality of rectangular openings **402** may abut the molded annular bead **328**. The second short side **504b** may face one of the plurality of square openings **404**. Each square opening of the plurality of square openings **404** may be defined by an opening on the first face **510** of the elastomeric seal **306**, an opening on the second face **610**, and a square surface **514**. Each square opening of the plurality of square openings **404** may include four perpendicularly arranged sides **512**. The plurality of square openings **404** and sides **512** may include a first side **512a** facing the outer circumference **414** and a second side **512b** facing one of the plurality of rectangular openings **402**.

[0053] The enlarged view **500** shows the plurality of first channels **406** and the plurality of second channels **408**. Each first channel of the plurality of first channels **406** may be defined by an opening on a first face **526** of the sealing ring **304**, an opening on an opposing, second face **528**, and a first channel surface **520**. Each first channel of the plurality of first channels **406** includes a pair of first channel long sides **518** and a pair of first channel short sides **516** arranged perpendicular to the pair of first channel long sides **518**. Each second channel of the plurality of second channels **408** may be defined by an opening on the first face **526** of the sealing ring **304**, an opening on the second face **528**, and a second channel surface **524**. Each second channel of the plurality of second channels **408** includes four perpendicularly arranged second channel sides **522**. The plurality of first channels **406** and the plurality of second channels **408** are shaped similarly and respectively aligned with the plurality of rectangular openings **402** and the plurality of square openings **404**. In this way, the sealing ring **304** and elastomeric seal **306** direct a flow of coolant from the plurality of coolant passages in the stator towards the plurality of end windings in the sealed cavity (e.g., from the coolant passages **206** to the end windings **228** in sealed cavity **226** in FIG. 2).

[0054] FIG. 6A and FIG. 6B shows a cross section view **600** of the elastomeric seal **306** of the sealing assembly **300** of a cooling system of an electric motor. The elastomeric seal **306** is shown in an uncompressed state. Elements introduced with reference to FIGS. 3-5 are numbered similarly and not reintroduced. A box **6-6** for an enlarged cross section view **650** depicted in FIG. 6B is provided in FIG. 6A.

[0055] Turning first to FIG. 6A, the cross section view **600** shows the molded annular bead **328** that defines the inner diameter **418** of the elastomeric seal **306**, and the planar portion **410** extending radially from the molded annular bead **328** to the outer diameter **416**. The elastomeric seal **306** may include a radial length **616** extending from the inner diameter **418** to the outer diameter **416**. In one example, the radial length may have a threshold range (e.g., a range of thresholds) from 31.5 mm to 37.5 mm. The elastomeric seal **306** further includes a wall **602**. In one example, in the uncompressed state shown by the cross section view **600** and the box **6-6**, the molded annular bead **328** includes a radial thickening **603** having a concave profile in cross-section. For example, the wall **602** at the radial thickening **603** may be relatively thicker than the wall **602** extending through the planar portion **410**. For example, the wall **602** may include a first axial length **612** that extends radially through the planar portion **410** and a second axial length **614** extending through the maximum dimension of the radial thickening **603** (e.g., along the y-axis). In one example, the second axial length **614** may include a threshold relative thickness range that is 1.7 to 1.5 times the first axial length **612**. In one example, the second axial length **614** may have a threshold range of 1.6 mm to 2.3 mm. In one example, first axial length may have a threshold range of 0.8 cm to 1.45 mm.

[0056] Turning to FIG. 6B, in one example, the molded annular bead **328** may include a first protrusion **606** arranged on the first face **510** of the elastomeric seal **306**, a second protrusion **608** arranged on a second face **610** of the elastomeric seal **306**, and a concave groove **604** arranged axially between the first protrusion **606** and the second protrusion **608**. The first protrusion **606** may include a first side **618** and second side **620**. The first protrusion **606** ramps down (e.g., relative to the y-axis) and inward toward the inner diameter **418** along the first side **618**. The first protrusion **606** ramps up and inward toward the inner diameter **418** along the second side **620**. The second protrusion **608** may include a third side **622** and fourth side **624**. The second protrusion **608** ramps up and inward toward the inner diameter **418** along the third side **622**. The second protrusion **608** ramps down and inward toward the inner diameter **418** along the fourth side **624**. The concave groove **604** may further include a first edge **630**, a second edge **632**, and a groove surface **634** extending axially therebetween. In one example, the first edge **630** may be approximately parallel with the first face **510** of the elastomeric seal **306** and the second edge **632** may approximately parallel with the second face **610**. In one example, the elastomeric seal may be bilaterally symmetric along an axis **628** of bilateral symmetry. The axis **628** of bilateral symmetry may be parallel with the Z-axis (e.g., the vertical axis).

[0057] FIG. 7 shows an enlarged view **700** of the sealing assembly **300** of a cooling system of an electric motor including the elastomeric seal **306** interposed radially between the lateral surface **302a** of the stator **302** and the sealing ring **304**. The enlarged view **700** depicts the sealing interface **326** formed between the first axial wall **322** of the sealing ring **304**, the second axial wall **324** of the stator **302**, and the elastomeric seal **306**. Elements introduced with reference to FIGS. 3-7 are numbered similarly and not reintroduced.

[0058] When the sealing assembly **300** is installed in the housing **110** of an electric motor assembly, upon final compression, a compressive force is exerted on the assembly. Arrows **702** indicate the compressive force exerted on the assembly formed between the sealing ring **304** and the stator **302**, and the elastomeric seal **306** arranged therebetween. When compressed axially, the elastomeric seal **306** may be compressed in a first direction **702a** by the sealing sleeve **308** and the inner flange **312**. Likewise, when compressed axially, the elastomeric seal **306** may be compressed in a second direction **702b** by the stator **302**. The first direction **702a** and the second direction **702b** are opposite to one another.

[0059] The molded annular bead **328** may be relatively thermoplastic. For example, under relatively low pressure conditions and/or heat (e.g., ambient), the molded annular bead **328** may have an axial dimension that is the second axial length **614**. However, under relatively higher pressure conditions and/or heat, such during final compression in the electric motor system, and

during motor operation, the molded annular bead **328** may change size. For example, upon final compression, the molded annular bead **328** may compresses axially to approximately the first axial length **612**. Further, axial compression of the molded annular bead **328** may cause extrusion (or extension) along the z-axis, filling the sealing interface **326**. For example, the molded annular bead may compress to a vertical length (e.g., along the z-axis) of approximately 2 mm. In this way, the molded annular bead **328** may sit on and form a seal with the minimal sealing land of the sealing interface **326**.

[0060] Due to the concave profile of the molded annular bead **328**, the molded annular bead **328** extends up to, and no further than, the first axial wall **322** of the sealing ring **304** in a radially inward direction, as indicated by arrow **704**, nor further than the second axial wall **324** that is approximately parallel with the first axial wall **322**. By expanding up to, and no further than, the first axial wall **322** of the sealing ring **304** and the second axial wall **324** of the stator **302**, the elastomeric seal **306** fluidly seals coolant in the sealed cavity **310** without protruding into the rotor core. Moreover, due to the relative plasticity of the elastomeric seal **306**, the elastomeric seal **306** may conform to the sealing interface **326** as pressure and heat conditions change during motor operation, which may cause expansion and contraction of the sealing interface **326**. The molded annular bead **328** may compress to the dimension of the sealing interface **326** under the range heat and pressure conditions of the operating motor without squeezing into the rotor core.

[0061] In this way, the disclosed sealing assembly for a cooling system for an electric motor may fluidly isolate coolant for stator end windings from an air gap formed between a stator and a rotor core. The sealing assembly includes an elastomeric seal configured for the stator end windings to pass through to a sealed cavity of the cooling system. The elastomeric seal further includes a molded annular bead that defines an inner diameter of the elastomeric seal, and which compresses axially, with respect to a longitudinal axis and/or rotational axis of the rotor, to seal an interface formed at the air gap. The molded annular bead may include a radial thickening having a concave profile in cross section. Upon final compression in an electric motor assembly, the geometry of the molded annular bead maintains the elastomeric seal on the minimal surface area, fluidly sealing the rotor without protruding past the interface. The technical effect of the sealing assembly is to effectively cool the stator end windings using a sealed cavity that reduces the chance of (e.g., avoids) coolant leakage into the rotor cavity, thereby increasing motor efficiency in relation to prior motor cooling systems.

[0062] The disclosure also provides support for a sealing assembly comprising: an elastomeric seal interposed radially between a lateral surface of a stator of an electric motor and a sealing ring, where the elastomeric seal comprises a plurality of openings configured to receive a plurality of end windings of the stator and a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate a coolant for the plurality of end windings from a rotor. In a first example of the system, in an uncompressed state the molded annular bead comprises a radial thickening having a concave profile in cross-section. In a second example of the system, optionally including the first example, in an uncompressed state the molded annular bead comprises a first protrusion arranged on a first face of the elastomeric seal, a second protrusion arranged on a second face of the elastomeric seal, and a concave groove arranged axially between the first protrusion and the second protrusion. In a third example of the system, optionally including one or both of the first and second examples, the first protrusion ramps down and inward toward the inner diameter from the first face of the elastomeric seal and the second protrusion ramps up and inward toward the inner diameter from the second face of the elastomeric seal. In a fourth example of the system, optionally including one or more or each of the first through third examples, the concave groove comprises a first edge, a second edge, and groove surface extending axially therebetween, wherein the first edge is approximately parallel with the first face of the elastomeric seal and the second edge is approximately parallel with the second face of the elastomeric seal. In a fifth example of the system, optionally including one or more or each

of the first through fourth examples, upon final compression in an electric motor assembly, the molded annular bead extends up to, and no further than, an axial wall of the sealing ring in a radially inward direction. In a sixth example of the system, optionally including one or more or each of the first through fifth examples, the elastomeric seal comprises a planar portion extending radially from the molded annular bead to an outer diameter. In a seventh example of the system, optionally including one or more or each of the first through sixth examples, the plurality of openings comprises a plurality of square openings arranged radially outward from a plurality of rectangular openings, wherein the plurality of rectangular openings are configured to receive the plurality of end windings of the stator, and the plurality of square openings are configured to be in fluid communication with the plurality of rectangular openings. In a eighth example of the system, optionally including one or more or each of the first through seventh examples, the elastomeric seal is formed from molded rubber and the plurality of rectangular openings and the plurality of square openings are die cut. In a ninth example of the system, optionally including one or more or each of the first through eighth examples, each square opening of the plurality of square openings is radially aligned with a rectangular opening of the plurality of rectangular openings with respect to a central axis of the elastomeric seal.

[0063] The disclosure also provides support for a cooling system for an electric motor comprising: a stator comprising a plurality of end windings and a plurality of coolant passages extending axially through the stator, the plurality of coolant passages in fluid communication with the plurality of end windings, a sealing ring coupled the stator and including a plurality of channels receiving the plurality of end windings, a sealing sleeve in contact with the sealing ring and forming a sealed cavity between the sealing sleeve and the sealing ring, the sealed cavity configured to at least partially to immerse the plurality of end windings at least partially in a coolant, and an elastomeric seal interposed radially between a lateral surface of the stator and the sealing ring, where the elastomeric seal comprises a plurality of openings configured to receive the plurality of end windings and a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate the coolant for the plurality of end windings from a rotor. In a first example of the system, the elastomeric seal extends radially from an inner circumferential section of the sealing ring to an outer circumferential section of the sealing ring, the inner circumferential section facing a rotor core and the outer circumferential section facing a motor housing. In a second example of the system, optionally including the first example, the plurality of channels direct a flow of coolant towards the plurality of end windings from the plurality of coolant passages in the stator. In a third example of the system, optionally including one or both of the first and second examples, in an uncompressed state the molded annular bead comprises a sealing surface having a concave profile in cross-section, and where upon final compression in an electric motor assembly, the molded annular bead extends up to, and no further than, an axial wall of the sealing ring in a radially inward direction. In a fourth example of the system, optionally including one or more or each of the first through third examples, the elastomeric seal comprises a plurality of square openings arranged radially outward from a plurality of rectangular openings, wherein the plurality of square openings are in fluid communication with the plurality of coolant passages in the stator.

[0064] The disclosure also provides support for a method for a cooling system comprising: flowing a coolant from a plurality of coolant passages that extend through a stator into a sealed cavity that is formed between an elastomeric seal, a sealing ring and a sealing sleeve and encloses end windings of the stator, wherein the elastomeric seal includes a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate the coolant from an air gap formed at an interface between the stator and a rotor. In a first example of the method, the sealed cavity is fluidly isolated from a rotor cavity. In a second example of the method, optionally including the first example, in an uncompressed state the molded annular bead comprises a sealing surface having a concave profile in cross-section, and

where upon final compression in an electric motor assembly, the molded annular bead extends up to, and no further than, an axial wall of the sealing ring in a radially inward direction. In a third example of the method, optionally including one or both of the first and second examples, the elastomeric seal comprises a planar portion extending radially from the molded annular bead to an outer diameter. In a fourth example of the method, optionally including one or more or each of the first through third examples, the planar portion comprises a plurality of rectangular openings and a plurality of square openings arranged radially outward from the plurality of rectangular openings, wherein the plurality of rectangular openings abut the molded annular bead, wherein the plurality of rectangular openings receive the end windings of the stator and the plurality of square openings flow coolant to the plurality of coolant passages.

[0065] In another representation, an electric motor comprising: a rotor, a stator circumferentially surrounding the rotor, an air gap positioned between the rotor and the stator, and an elastomeric seal interposed radially between a lateral surface of the stator and a sealing ring, where the elastomeric seal comprises a plurality of openings configured to receive a plurality of end windings of the stator and a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate a coolant for the plurality of end windings from entering the air gap.

[0066] FIGS. 1-7 are drawn approximately to scale, aside from the schematically depicted components. However, the components may have other relative dimensions, in other embodiments.

[0067] FIGS. 1-7 show example configurations with relative positioning of the various components. If shown directly contacting each other, or directly coupled, then such elements may be referred to as directly contacting or directly coupled, respectively, at least in one example. Similarly, elements shown contiguous or adjacent to one another may be contiguous or adjacent to each other, respectively, at least in one example. As an example, components laying in face-sharing contact with each other may be referred to as in face-sharing contact. As another example, elements positioned apart from each other with only a space there-between and no other components may be referred to as such, in at least one example. As yet another example, elements shown above/below one another, at opposite sides to one another, or to the left/right of one another may be referred to as such, relative to one another. Further, as shown in the figures, a topmost element or point of element may be referred to as a “top” of the component and a bottommost element or point of the element may be referred to as a “bottom” of the component, in at least one example. As used herein, top/bottom, upper/lower, above/below, may be relative to a vertical axis of the figures and used to describe positioning of elements of the figures relative to one another. As such, elements shown above other elements are positioned vertically above the other elements, in one example. As yet another example, shapes of the elements depicted within the figures may be referred to as having those shapes (e.g., such as being circular, straight, planar, curved, rounded, chamfered, angled, or the like). Additionally, elements co-axial with one another may be referred to as such, in one example. Further, elements shown intersecting one another may be referred to as intersecting elements or intersecting one another, in at least one example. Further still, an element shown within another element or shown outside of another element may be referred to as such, in one example. In other examples, elements offset from one another may be referred to as such. Even further, elements which are coaxial or parallel to one another may be referred to as such. As yet another example, features described as “substantially” shaped, e.g., annular, flat, planar, prismatic, circular, etc., means that the features are sufficiently shaped as such to be considered having the shape by one skilled in the art.

[0068] Note that the example control and estimation routines included herein can be used with various motor configurations. The control methods and routines disclosed herein may be stored as executable instructions in non-transitory memory and may be carried out by the control system including the controller in combination with the various sensors, actuators, and other electric drive and/or vehicle hardware in combination with the electronic controller. As such, the described

actions, operations, and/or functions may graphically represent code to be programmed into non-transitory memory of the computer readable storage medium in the vehicle and/or driveline control system. One or more of the illustrated actions, operations and/or functions may be repeatedly performed depending on the particular strategy being used. One or more of the method steps described herein may be omitted if desired.

[0069] While various embodiments have been described above, it should be understood that they have been presented by way of example, and not limitation. It will be apparent to persons skilled in the relevant arts that the disclosed subject matter may be embodied in other specific forms without departing from the spirit of the subject matter. The embodiments described above are therefore to be considered in all respects as illustrative, not restrictive. As such, the configurations and routines disclosed herein are exemplary in nature, and that these specific examples are not to be considered in a limiting sense, because numerous variations are possible. The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various systems and configurations, and other features, functions, and/or properties disclosed herein.

[0070] The following claims particularly point out certain combinations and sub-combinations regarded as novel and non-obvious. These claims may refer to “an” element or “a first” element or the equivalent thereof. Such claims should be understood to include incorporation of one or more such elements, neither requiring nor excluding two or more such elements. Other combinations and sub-combinations of the disclosed features, functions, elements, and/or properties may be claimed through amendment of the present claims or through presentation of new claims in this or a related application. Such claims, whether broader, narrower, equal, or different in scope to the original claims, also are regarded as included within the subject matter of the present disclosure.

Claims

1. A sealing assembly comprising: an elastomeric seal interposed radially between a lateral surface of a stator of an electric motor and a sealing ring, where the elastomeric seal comprises a plurality of openings configured to receive a plurality of end windings of the stator and a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate a coolant for the plurality of end windings from a rotor.
2. The sealing assembly of claim 1, where in an uncompressed state the molded annular bead comprises a radial thickening having a concave profile in cross-section.
3. The sealing assembly of claim 1, where in an uncompressed state the molded annular bead comprises a first protrusion arranged on a first face of the elastomeric seal, a second protrusion arranged on a second face of the elastomeric seal, and a concave groove arranged axially between the first protrusion and the second protrusion.
4. The sealing assembly of claim 3, wherein the first protrusion ramps down and inward toward the inner diameter from the first face of the elastomeric seal and the second protrusion ramps up and inward toward the inner diameter from the second face of the elastomeric seal.
5. The sealing assembly of claim 3, wherein the concave groove comprises a first edge, a second edge, and groove surface extending axially therebetween, wherein the first edge is approximately parallel with the first face of the elastomeric seal and the second edge is approximately parallel with the second face of the elastomeric seal.
6. The sealing assembly of claim 1, where upon final compression in an electric motor assembly, the molded annular bead extends up to, and no further than, an axial wall of the sealing ring in a radially inward direction.
7. The sealing assembly of claim 1, wherein the elastomeric seal comprises a planar portion extending radially from the molded annular bead to an outer diameter.
8. The sealing assembly of claim 1, wherein the plurality of openings comprises a plurality of square openings arranged radially outward from a plurality of rectangular openings, wherein the

plurality of rectangular openings are configured to receive the plurality of end windings of the stator, and the plurality of square openings are configured to be in fluid communication with the plurality of rectangular openings.

9. The sealing assembly of claim 8, wherein the elastomeric seal is formed from molded rubber and the plurality of rectangular openings and the plurality of square openings are die cut.

10. The sealing assembly of claim 8, wherein each square opening of the plurality of square openings is radially aligned with a rectangular opening of the plurality of rectangular openings with respect to a central axis of the elastomeric seal.

11. A cooling system for an electric motor comprising: a stator comprising a plurality of end windings and a plurality of coolant passages extending axially through the stator, the plurality of coolant passages in fluid communication with the plurality of end windings; a sealing ring coupled to the stator and including a plurality of channels receiving the plurality of end windings; a sealing sleeve in contact with the sealing ring and forming a sealed cavity between the sealing sleeve and the sealing ring, the sealed cavity configured to at least partially immerse the plurality of end windings at least partially in a coolant; and an elastomeric seal interposed radially between a lateral surface of the stator and the sealing ring, where the elastomeric seal comprises a plurality of openings configured to receive the plurality of end windings and a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate the coolant for the plurality of end windings from a rotor.

12. The cooling system of claim 11, wherein the elastomeric seal extends radially from an inner circumferential section of the sealing ring to an outer circumferential section of the sealing ring, the inner circumferential section facing a rotor core and the outer circumferential section facing a motor housing.

13. The cooling system of claim 11, wherein the plurality of channels direct a flow of coolant towards the plurality of end windings from the plurality of coolant passages in the stator.

14. The cooling system of claim 11, where in an uncompressed state the molded annular bead comprises a sealing surface having a concave profile in cross-section, and where upon final compression in an electric motor assembly, the molded annular bead extends up to, and no further than, an axial wall of the sealing ring in a radially inward direction.

15. The cooling system of claim 11, wherein the elastomeric seal comprises a plurality of square openings arranged radially outward from a plurality of rectangular openings, wherein the plurality of square openings are in fluid communication with the plurality of coolant passages in the stator.

16. A method for a cooling system comprising: flowing a coolant from a plurality of coolant passages that extend through a stator into a sealed cavity that is formed between an elastomeric seal, a sealing ring and a sealing sleeve and encloses end windings of the stator, wherein the elastomeric seal includes a molded annular bead arranged to define an inner diameter of the elastomeric seal and configured to be compressed axially to further separate the coolant from an air gap formed at an interface between the stator and a rotor.

17. The method of claim 16, wherein the sealed cavity is fluidly isolated from a rotor cavity.

18. The method of claim 16, where in an uncompressed state the molded annular bead comprises a sealing surface having a concave profile in cross-section, and where upon final compression in an electric motor assembly, the molded annular bead extends up to, and no further than, an axial wall of the sealing ring in a radially inward direction.

19. The method of claim 16, wherein the elastomeric seal comprises a planar portion extending radially from the molded annular bead to an outer diameter.

20. The method of claim 19, wherein the planar portion comprises a plurality of rectangular openings and a plurality of square openings arranged radially outward from the plurality of rectangular openings, wherein the plurality of rectangular openings abut the molded annular bead, wherein the plurality of rectangular openings receive the end windings of the stator and the plurality of square openings flow coolant to the plurality of coolant passages.

